Average rates of change

1

a -4 b -3 c -7 d Variable

2 Constant

3

a -1 b -1 c -1 d Linear

4

a -4 b -4 c -4 d Linear

5

a Constant b Variable c Constant d Variable

e Variable f Variable g Variable h Variable

6

a $\frac{23}{4}$ b 5 c $\frac{23}{5}$ d Non-linear

7 Non-linear

8

a 1

b 2

c 4

d

Interval	$[3,4]$	$[4,5]$	$[5,6]$
Average rate of change	8	16	32

9

a $\frac{9}{10}$

b The cost of a call increases by $\$0.90$ for each additional minute.

10

a 500

b Aside from the pre-ordered tickets, there were 500 tickets purchased each day.

11

a $-\frac{1}{250}$

b The price of each toy decreases by $\$1.00$ for each additional 250 units in supply.

12

a 0

b The number of earthquakes per year is a constant value.

Applications

13

a $y = 11$

b $y = 26$

c $\frac{15}{4}$

14 272

15 $\$300$ per year

16 60

17 6.5 insects per week

18 1.6

19

a

i 9 ft/s

ii 3 ft/s

iii -3 ft/s

iv -9 ft/s

b Between $t = 0$ and $t = 1$, and between $t = 3$ and $t = 4$

c The ball is falling down.

20

a

Year	0	1	2	3	4	5
Tobia's staff	310	269	228	187	146	105
Gwen's staff	310	302	286	262	230	190

b -41

c -32

d Linear, because it decreases by the same amount each year.

e Non-linear, because it decreases by more each year.

f Tobias's

g Gwen's

21

a

Month	0	1	2	3	4	5
Mario's clients	0	27	54	81	108	135
Christa's clients	0	6	18	36	60	90

b Linear

c Non-linear

d 27

e 18

f Mario's

g Christa's

22

a

Month	0	1	2	3	4	5
Value of Harry's investment	2000	2060	2120	2180	2240	2300
Value of Skye's investment	2000	2040	2080.8	2122.42	2164.87	2208.17

b $60

c $40.8

d Linear

e Non-linear

23

a $y = 4x + 20$

b $y = 20\,(1.10)^x$

c 4 turbines each year

d 9.59 turbines each year

e Option A